PEMS03

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	14.42 ± 0.21	13.70 ± 0.12	15.14 ± 0.10	14.22 ± 0.29	30.61 ± 1.22	13.09 ± 0.16	16.25 ± 0.19	12.76 ± 0.05	13.22 ± 0.11	13.88 ± 0.85	13.42 ± 0.27	14.18 ± 0.31	13.10 ± 0.12	12.89 ± 0.15	14.28 ± 0.15	16.46 ± 0.42	13.16 ± 0.13	14.17 ± 0.07	12.67	12.53	11.73 ± 0.01	13.20	12.13 ± 0.01	14.50	16.03 ± 0.64	13.66	12.16 ± 0.00	11.81 ± 0.11	**10.57 ± 0.03**	*11.18 ± 0.03*
	6	14.67 ± 0.18	14.89 ± 0.10	15.63 ± 0.10	15.08 ± 0.23	31.21 ± 1.19	14.01 ± 0.14	17.32 ± 0.16	13.68 ± 0.05	14.17 ± 0.17	14.76 ± 0.81	14.28 ± 0.20	15.64 ± 0.48	14.00 ± 0.12	13.82 ± 0.13	14.89 ± 0.14	17.86 ± 0.45	14.05 ± 0.14	15.45 ± 0.06	14.85	14.74	13.09 ± 0.01	15.55	13.87 ± 0.01	17.22	18.39 ± 0.59	15.87	13.70 ± 0.01	12.22 ± 0.10	**11.08 ± 0.03**	*12.10 ± 0.04*
	12	16.27 ± 0.18	17.57 ± 0.08	17.27 ± 0.11	17.24 ± 0.30	32.66 ± 0.99	15.92 ± 0.14	19.41 ± 0.20	15.46 ± 0.09	16.10 ± 0.35	16.57 ± 0.78	16.24 ± 0.18	18.65 ± 0.79	15.86 ± 0.15	15.74 ± 0.13	16.71 ± 0.13	21.02 ± 0.59	15.95 ± 0.13	18.34 ± 0.04	20.14	19.88	15.90 ± 0.02	20.21	17.38 ± 0.02	22.10	22.80 ± 0.59	20.35	16.86 ± 0.01	*12.98 ± 0.13*	**11.93 ± 0.05**	13.74 ± 0.07
	Avg	14.99 ± 0.18	15.15 ± 0.10	15.88 ± 0.10	15.33 ± 0.26	31.37 ± 1.11	14.16 ± 0.15	17.47 ± 0.16	13.80 ± 0.05	14.32 ± 0.20	14.90 ± 0.81	14.47 ± 0.22	15.91 ± 0.48	14.15 ± 0.13	13.98 ± 0.14	15.14 ± 0.14	18.20 ± 0.47	14.21 ± 0.13	15.73 ± 0.06	15.45	15.27	13.31 ± 0.01	15.91	14.16 ± 0.01	17.59	18.68 ± 0.59	16.19	13.95 ± 0.01	12.26 ± 0.11	**11.10 ± 0.04**	*12.19 ± 0.04*
RMSE	3	23.64 ± 0.22	22.30 ± 0.20	28.88 ± 0.42	23.01 ± 0.41	39.61 ± 1.17	21.27 ± 0.15	25.71 ± 0.14	20.83 ± 0.06	21.39 ± 0.10	22.19 ± 1.02	21.71 ± 0.38	22.90 ± 0.58	21.23 ± 0.22	20.98 ± 0.13	25.67 ± 0.42	25.72 ± 0.59	21.37 ± 0.14	23.03 ± 0.04	21.63	21.37	20.23 ± 0.02	22.46	20.93 ± 0.03	25.10	27.89 ± 1.02	23.21	20.77 ± 0.02	19.61 ± 0.15	**17.79 ± 0.06**	*18.79 ± 0.05*
	6	24.33 ± 0.17	24.71 ± 0.16	29.72 ± 0.40	24.70 ± 0.33	41.30 ± 1.09	23.20 ± 0.12	28.00 ± 0.10	22.70 ± 0.10	23.35 ± 0.22	24.08 ± 0.95	23.56 ± 0.28	25.90 ± 0.88	23.13 ± 0.21	22.93 ± 0.11	26.68 ± 0.38	27.99 ± 0.61	23.14 ± 0.15	25.74 ± 0.04	26.10	25.87	23.16 ± 0.04	27.14	24.74 ± 0.04	30.33	32.52 ± 0.89	27.41	23.96 ± 0.03	*20.52 ± 0.12*	**18.88 ± 0.07**	20.62 ± 0.07
	12	27.56 ± 0.21	29.82 ± 0.14	32.46 ± 0.37	28.79 ± 0.51	44.99 ± 0.98	26.94 ± 0.16	32.37 ± 0.35	26.14 ± 0.20	27.11 ± 0.58	27.75 ± 0.94	27.50 ± 0.37	31.87 ± 1.46	26.81 ± 0.26	26.70 ± 0.15	29.78 ± 0.35	33.28 ± 0.82	26.74 ± 0.15	31.45 ± 0.04	37.11	36.36	29.28 ± 0.07	36.42	32.31 ± 0.09	39.29	40.97 ± 0.93	35.98	30.56 ± 0.04	*22.05 ± 0.13*	**20.50 ± 0.08**	23.65 ± 0.12
	Avg	24.90 ± 0.19	25.15 ± 0.17	30.12 ± 0.40	25.15 ± 0.39	41.64 ± 0.99	23.46 ± 0.14	28.29 ± 0.08	22.89 ± 0.11	23.60 ± 0.28	24.32 ± 0.98	23.89 ± 0.34	26.40 ± 0.88	23.37 ± 0.22	23.18 ± 0.13	27.10 ± 0.38	28.58 ± 0.65	23.42 ± 0.14	26.23 ± 0.04	27.38	26.98	23.66 ± 0.03	27.86	25.36 ± 0.04	30.92	32.99 ± 0.89	28.00	24.51 ± 0.03	*20.58 ± 0.12*	**18.87 ± 0.07**	20.72 ± 0.07
MAPE (%)	3	28.08 ± 0.88	24.72 ± 0.40	28.13 ± 0.82	26.65 ± 0.67	50.48 ± 2.93	25.11 ± 0.64	28.43 ± 0.78	24.05 ± 0.22	26.49 ± 1.13	26.87 ± 2.27	26.48 ± 0.70	27.89 ± 1.36	25.03 ± 0.27	24.96 ± 0.64	25.31 ± 0.62	49.62 ± 2.31	25.56 ± 0.78	26.28 ± 0.91	22.73	22.59	21.07 ± 0.03	23.53	21.74 ± 0.02	23.11	23.82 ± 0.36	23.23	22.00 ± 0.01	22.52 ± 0.64	**20.86 ± 0.18**	*22.34 ± 0.26*
	6	28.30 ± 0.86	26.15 ± 0.42	28.46 ± 0.91	27.65 ± 0.69	49.81 ± 2.94	26.03 ± 0.57	29.35 ± 0.93	24.75 ± 0.17	27.29 ± 1.48	27.58 ± 2.27	27.28 ± 0.67	29.56 ± 0.89	25.82 ± 0.29	25.89 ± 0.57	26.15 ± 0.61	52.24 ± 2.52	26.53 ± 0.74	27.99 ± 0.96	25.08	25.06	22.52 ± 0.03	26.19	23.46 ± 0.02	25.83	26.96 ± 0.32	25.68	23.83 ± 0.01	*22.95 ± 0.64*	**21.37 ± 0.18**	23.27 ± 0.26
	12	30.21 ± 0.81	29.78 ± 0.50	30.89 ± 0.96	30.44 ± 0.82	49.61 ± 2.26	28.34 ± 0.49	31.64 ± 1.66	26.62 ± 0.25	29.70 ± 2.58	29.44 ± 2.39	29.57 ± 0.77	34.17 ± 1.00	28.11 ± 0.57	28.20 ± 0.48	28.48 ± 0.63	58.49 ± 3.04	28.91 ± 0.60	32.53 ± 1.08	31.38	31.43	25.67 ± 0.02	31.92	27.12 ± 0.01	31.32	32.00 ± 0.24	30.84	27.77 ± 0.03	*24.00 ± 0.63*	**22.44 ± 0.20**	25.20 ± 0.30
	Avg	28.69 ± 0.86	26.56 ± 0.43	28.95 ± 0.89	28.01 ± 0.71	50.04 ± 2.80	26.29 ± 0.58	29.62 ± 1.04	24.96 ± 0.21	27.63 ± 1.67	27.78 ± 2.30	27.58 ± 0.64	30.12 ± 0.81	26.12 ± 0.31	26.15 ± 0.57	26.44 ± 0.62	52.94 ± 2.57	26.79 ± 0.71	28.55 ± 0.98	25.88	25.83	22.80 ± 0.03	26.71	23.79 ± 0.01	26.24	27.23 ± 0.31	26.15	24.18 ± 0.01	*23.06 ± 0.64*	**21.46 ± 0.18**	23.43 ± 0.27

PEMS04

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	21.15 ± 0.32	20.23 ± 0.12	22.27 ± 0.38	22.00 ± 0.32	153.31 ± 6.85	19.92 ± 0.23	36.81 ± 0.33	19.34 ± 0.22	19.74 ± 0.48	23.62 ± 3.33	21.20 ± 1.34	31.75 ± 9.85	20.31 ± 0.70	19.59 ± 0.23	21.19 ± 0.21	22.37 ± 0.62	19.59 ± 0.15	21.27 ± 0.23	17.83	17.60	16.52 ± 0.03	19.32	17.78 ± 0.02	21.81	26.08 ± 0.33	20.63	17.86 ± 0.03	18.62 ± 0.99	**15.81 ± 0.07**	*16.56 ± 0.04*
	6	21.75 ± 0.26	22.06 ± 0.11	23.22 ± 0.37	22.97 ± 0.28	153.21 ± 6.57	21.35 ± 0.19	38.21 ± 0.32	20.78 ± 0.26	21.18 ± 0.50	24.90 ± 3.16	22.41 ± 1.06	33.61 ± 9.31	21.63 ± 0.64	21.04 ± 0.19	21.96 ± 0.17	23.63 ± 0.63	20.94 ± 0.11	23.12 ± 0.17	20.98	20.75	18.59 ± 0.04	22.92	20.46 ± 0.02	26.14	30.01 ± 0.17	24.04	20.27 ± 0.03	19.33 ± 0.95	**16.65 ± 0.11**	*18.01 ± 0.04*
	12	24.38 ± 0.25	26.11 ± 0.06	25.97 ± 0.36	26.06 ± 0.26	153.51 ± 6.60	24.45 ± 0.19	41.12 ± 0.17	23.68 ± 0.33	24.16 ± 0.63	27.74 ± 2.99	25.55 ± 1.01	37.67 ± 7.70	24.64 ± 0.65	24.17 ± 0.18	24.86 ± 0.16	26.74 ± 0.68	23.91 ± 0.11	27.45 ± 0.13	28.71	28.27	22.84 ± 0.07	30.08	25.86 ± 0.01	33.86	39.47 ± 0.65	30.91	25.23 ± 0.02	*20.61 ± 0.94*	**18.06 ± 0.18**	20.62 ± 0.08
	Avg	22.22 ± 0.27	22.45 ± 0.09	23.60 ± 0.37	23.41 ± 0.27	153.33 ± 6.62	21.64 ± 0.20	38.46 ± 0.27	21.01 ± 0.26	21.43 ± 0.53	25.16 ± 3.18	22.78 ± 1.14	34.02 ± 9.03	21.93 ± 0.67	21.33 ± 0.20	22.43 ± 0.18	23.99 ± 0.62	21.22 ± 0.11	23.58 ± 0.18	21.88	21.58	18.94 ± 0.04	23.49	20.92 ± 0.02	26.72	31.16 ± 0.33	24.52	20.68 ± 0.03	19.38 ± 0.94	**16.70 ± 0.11**	*18.15 ± 0.05*
RMSE	3	32.49 ± 0.37	31.23 ± 0.11	36.57 ± 0.57	33.27 ± 0.26	185.75 ± 7.85	30.57 ± 0.22	48.99 ± 0.57	29.87 ± 0.23	30.41 ± 0.49	36.55 ± 4.40	31.87 ± 1.33	44.57 ± 11.20	30.86 ± 0.67	30.17 ± 0.22	33.49 ± 0.37	34.23 ± 0.76	30.24 ± 0.20	32.51 ± 0.25	29.20	28.79	27.45 ± 0.07	30.99	29.58 ± 0.02	34.99	41.11 ± 0.57	32.99	29.40 ± 0.11	29.26 ± 1.26	**25.94 ± 0.11**	*27.03 ± 0.04*
	6	33.93 ± 0.28	34.42 ± 0.10	38.28 ± 0.55	35.06 ± 0.24	186.05 ± 7.49	33.18 ± 0.20	51.49 ± 0.51	32.43 ± 0.30	33.00 ± 0.54	38.80 ± 4.09	34.23 ± 1.05	47.58 ± 10.31	33.36 ± 0.61	32.80 ± 0.20	34.98 ± 0.29	36.16 ± 0.79	32.65 ± 0.16	35.90 ± 0.21	34.20	33.78	31.20 ± 0.10	36.72	34.39 ± 0.01	41.80	47.11 ± 0.31	38.04	33.63 ± 0.09	30.61 ± 1.20	**27.44 ± 0.14**	*29.43 ± 0.07*
	12	38.46 ± 0.30	40.98 ± 0.17	42.75 ± 0.53	40.07 ± 0.30	187.60 ± 7.68	38.43 ± 0.24	56.38 ± 0.30	37.23 ± 0.45	37.99 ± 0.82	43.47 ± 3.76	39.70 ± 1.19	53.93 ± 7.70	38.50 ± 0.67	38.08 ± 0.25	39.67 ± 0.26	40.75 ± 0.93	37.57 ± 0.19	43.10 ± 0.21	47.13	46.16	39.07 ± 0.21	48.32	43.96 ± 0.03	53.81	64.62 ± 2.50	48.84	42.50 ± 0.06	*32.76 ± 1.20*	**29.73 ± 0.25**	33.51 ± 0.13
	Avg	34.58 ± 0.32	34.96 ± 0.12	38.82 ± 0.55	35.69 ± 0.22	186.37 ± 7.58	33.58 ± 0.22	51.85 ± 0.46	32.73 ± 0.31	33.35 ± 0.60	39.17 ± 4.12	34.78 ± 1.18	48.20 ± 9.87	33.78 ± 0.64	33.21 ± 0.22	35.63 ± 0.31	36.65 ± 0.80	33.04 ± 0.17	36.54 ± 0.22	35.80	35.22	31.89 ± 0.12	37.69	35.20 ± 0.02	42.74	49.27 ± 0.58	38.87	34.42 ± 0.09	30.64 ± 1.20	**27.46 ± 0.15**	*29.59 ± 0.07*
MAPE (%)	3	19.95 ± 0.41	17.57 ± 0.43	19.35 ± 0.26	18.96 ± 0.52	82.38 ± 3.00	17.87 ± 0.35	25.59 ± 0.52	16.67 ± 0.26	17.63 ± 0.50	21.16 ± 3.83	20.61 ± 1.82	37.87 ± 18.81	18.37 ± 0.71	17.74 ± 0.35	17.69 ± 0.30	26.18 ± 1.42	18.27 ± 0.21	17.68 ± 0.31	14.82	14.71	13.71 ± 0.01	16.27	14.69 ± 0.02	16.45	17.91 ± 0.07	16.23	14.90 ± 0.01	16.44 ± 0.94	**14.27 ± 0.03**	*15.20 ± 0.22*
	6	20.17 ± 0.42	18.89 ± 0.42	19.76 ± 0.28	19.60 ± 0.54	82.26 ± 3.00	18.86 ± 0.28	26.67 ± 0.66	17.58 ± 0.32	18.53 ± 0.54	22.08 ± 3.77	21.21 ± 1.50	38.74 ± 17.46	19.14 ± 0.60	18.75 ± 0.28	18.18 ± 0.26	27.56 ± 1.51	19.11 ± 0.24	19.09 ± 0.32	17.01	16.93	15.00 ± 0.01	18.84	16.33 ± 0.02	19.03	20.80 ± 0.03	18.75	16.48 ± 0.02	16.86 ± 0.94	**14.81 ± 0.04**	*16.25 ± 0.28*
	12	21.91 ± 0.40	22.29 ± 0.42	21.86 ± 0.33	21.79 ± 0.59	82.01 ± 3.01	21.34 ± 0.25	29.13 ± 1.08	19.79 ± 0.43	20.92 ± 0.83	24.23 ± 3.51	23.44 ± 1.32	42.12 ± 15.90	21.54 ± 0.39	21.24 ± 0.24	20.14 ± 0.21	31.05 ± 1.68	21.23 ± 0.34	22.99 ± 0.46	22.74	22.62	17.78 ± 0.01	24.24	19.78 ± 0.02	23.86	26.17 ± 0.16	23.60	19.89 ± 0.03	*17.90 ± 1.02*	**15.90 ± 0.09**	18.36 ± 0.38
	Avg	20.53 ± 0.40	19.29 ± 0.42	20.15 ± 0.28	19.94 ± 0.55	82.25 ± 2.99	19.15 ± 0.30	26.93 ± 0.71	17.82 ± 0.31	18.83 ± 0.59	22.29 ± 3.69	21.54 ± 1.53	39.19 ± 17.54	19.49 ± 0.57	19.04 ± 0.29	18.51 ± 0.26	27.98 ± 1.51	19.36 ± 0.25	19.60 ± 0.33	17.72	17.62	15.25 ± 0.01	19.32	16.65 ± 0.02	19.38	21.25 ± 0.07	19.11	16.78 ± 0.02	16.97 ± 0.95	**14.89 ± 0.05**	*16.41 ± 0.28*

PEMS05

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	11.22 ± 0.09	10.77 ± 0.03	11.31 ± 0.11	11.71 ± 0.12	24.31 ± 1.08	10.54 ± 0.04	14.56 ± 0.51	10.39 ± 0.07	10.54 ± 0.12	10.93 ± 0.34	10.73 ± 0.17	15.58 ± 3.35	10.58 ± 0.05	10.45 ± 0.04	11.03 ± 0.07	11.98 ± 0.14	10.69 ± 0.07	11.00 ± 0.06	10.86	10.72	10.01 ± 0.01	11.20	10.44 ± 0.01	12.13	13.76 ± 0.38	11.56	10.39 ± 0.00	9.88 ± 0.06	**9.20 ± 0.02**	*9.63 ± 0.01*
	6	11.49 ± 0.06	11.55 ± 0.04	11.79 ± 0.09	12.34 ± 0.13	24.77 ± 0.99	11.20 ± 0.04	15.76 ± 0.54	11.00 ± 0.07	11.21 ± 0.12	11.55 ± 0.31	11.36 ± 0.14	16.21 ± 3.54	11.21 ± 0.03	11.11 ± 0.04	11.48 ± 0.05	12.44 ± 0.11	11.30 ± 0.06	11.96 ± 0.05	12.33	12.23	10.94 ± 0.01	12.84	11.62 ± 0.01	14.04	15.48 ± 0.25	12.97	11.46 ± 0.00	*10.12 ± 0.07*	**9.50 ± 0.02**	10.21 ± 0.03
	12	12.63 ± 0.05	13.21 ± 0.05	13.04 ± 0.08	14.10 ± 0.18	26.01 ± 1.05	12.57 ± 0.06	18.26 ± 0.54	12.24 ± 0.11	12.57 ± 0.16	12.87 ± 0.27	12.72 ± 0.17	17.64 ± 3.79	12.55 ± 0.05	12.48 ± 0.06	12.72 ± 0.05	13.56 ± 0.08	12.62 ± 0.05	14.06 ± 0.06	16.08	15.95	12.93 ± 0.02	16.14	14.09 ± 0.01	17.55	18.96 ± 0.26	16.18	13.73 ± 0.00	*10.60 ± 0.08*	**10.01 ± 0.04**	11.28 ± 0.05
	Avg	11.69 ± 0.07	11.69 ± 0.04	11.94 ± 0.10	12.57 ± 0.13	24.94 ± 0.99	11.31 ± 0.04	15.98 ± 0.52	11.10 ± 0.07	11.32 ± 0.13	11.67 ± 0.31	11.48 ± 0.16	16.35 ± 3.53	11.33 ± 0.04	11.22 ± 0.04	11.64 ± 0.06	12.57 ± 0.10	11.42 ± 0.06	12.16 ± 0.05	12.79	12.67	11.12 ± 0.01	13.11	11.85 ± 0.01	14.34	15.75 ± 0.28	13.26	11.66 ± 0.00	*10.16 ± 0.07*	**9.52 ± 0.02**	10.27 ± 0.03
RMSE	3	17.36 ± 0.11	16.92 ± 0.07	18.29 ± 0.18	17.96 ± 0.15	33.60 ± 1.18	16.59 ± 0.07	23.55 ± 0.72	16.33 ± 0.09	16.61 ± 0.11	17.14 ± 0.48	16.84 ± 0.20	22.45 ± 3.39	16.58 ± 0.06	16.46 ± 0.06	17.64 ± 0.10	18.03 ± 0.22	16.69 ± 0.08	17.33 ± 0.06	17.39	17.14	16.31 ± 0.01	17.78	16.98 ± 0.00	19.54	21.76 ± 0.54	18.39	16.77 ± 0.01	15.54 ± 0.09	**14.67 ± 0.03**	*15.41 ± 0.03*
	6	18.03 ± 0.07	18.31 ± 0.07	19.20 ± 0.16	19.12 ± 0.18	34.52 ± 1.07	17.84 ± 0.06	25.94 ± 0.75	17.49 ± 0.13	17.87 ± 0.12	18.33 ± 0.43	18.07 ± 0.18	23.58 ± 3.68	17.79 ± 0.05	17.71 ± 0.06	18.43 ± 0.07	18.81 ± 0.16	17.86 ± 0.07	19.09 ± 0.05	19.96	19.78	18.19 ± 0.01	20.61	19.44 ± 0.01	22.68	24.70 ± 0.40	20.68	18.85 ± 0.01	*16.02 ± 0.10*	**15.24 ± 0.04**	16.48 ± 0.05
	12	20.16 ± 0.08	21.07 ± 0.10	21.40 ± 0.15	22.14 ± 0.27	36.72 ± 1.19	20.30 ± 0.12	30.76 ± 0.61	19.68 ± 0.22	20.32 ± 0.22	20.68 ± 0.38	20.57 ± 0.27	25.93 ± 3.97	20.19 ± 0.09	20.17 ± 0.12	20.58 ± 0.06	20.61 ± 0.16	20.20 ± 0.08	22.73 ± 0.08	26.60	26.27	22.43 ± 0.04	26.26	24.56 ± 0.02	28.35	30.31 ± 0.41	25.89	23.50 ± 0.02	*16.89 ± 0.13*	**16.13 ± 0.06**	18.33 ± 0.10
	Avg	18.34 ± 0.09	18.52 ± 0.08	19.43 ± 0.16	19.49 ± 0.18	34.77 ± 1.06	18.03 ± 0.07	26.35 ± 0.67	17.63 ± 0.13	18.04 ± 0.14	18.50 ± 0.44	18.26 ± 0.21	23.77 ± 3.62	17.97 ± 0.04	17.89 ± 0.07	18.69 ± 0.08	19.00 ± 0.17	18.04 ± 0.07	19.40 ± 0.06	20.79	20.54	18.61 ± 0.02	21.07	19.92 ± 0.01	23.12	25.07 ± 0.41	21.14	19.31 ± 0.00	*16.07 ± 0.10*	**15.25 ± 0.04**	16.55 ± 0.06
MAPE (%)	3	26.28 ± 0.86	23.31 ± 0.62	25.81 ± 0.68	25.60 ± 0.88	114.78 ± 6.27	23.06 ± 0.42	27.15 ± 0.88	22.28 ± 0.43	23.18 ± 0.88	23.88 ± 0.92	23.69 ± 0.81	59.98 ± 37.51	23.90 ± 0.94	22.97 ± 0.42	23.40 ± 0.55	33.91 ± 0.52	23.53 ± 0.44	23.50 ± 0.68	22.07	21.96	20.12 ± 0.01	23.23	20.62 ± 0.02	22.62	23.96 ± 0.28	22.69	21.26 ± 0.01	22.26 ± 0.46	**19.94 ± 0.11**	*20.14 ± 0.06*
	6	26.59 ± 0.75	24.92 ± 0.62	26.46 ± 0.61	26.73 ± 0.88	115.48 ± 6.90	24.15 ± 0.42	28.67 ± 0.81	23.24 ± 0.42	24.29 ± 0.95	24.84 ± 0.87	24.82 ± 0.74	60.44 ± 37.27	24.93 ± 0.92	24.05 ± 0.42	24.20 ± 0.51	35.01 ± 0.59	24.65 ± 0.36	25.54 ± 0.69	24.65	24.72	21.39 ± 0.01	26.15	22.15 ± 0.01	25.53	27.63 ± 0.14	25.33	22.97 ± 0.01	22.67 ± 0.37	**20.45 ± 0.09**	*21.01 ± 0.02*
	12	28.62 ± 0.63	28.72 ± 0.60	28.84 ± 0.49	30.08 ± 0.80	118.51 ± 8.78	26.77 ± 0.42	32.35 ± 0.78	25.61 ± 0.55	26.89 ± 1.00	27.37 ± 0.73	27.67 ± 0.80	63.79 ± 37.30	27.61 ± 0.75	26.67 ± 0.42	26.50 ± 0.51	38.11 ± 0.99	27.28 ± 0.38	30.73 ± 0.73	31.79	32.01	24.30 ± 0.03	32.54	25.58 ± 0.01	31.30	33.24 ± 0.06	31.29	26.70 ± 0.01	23.86 ± 0.49	**21.51 ± 0.11**	*22.78 ± 0.04*
	Avg	26.98 ± 0.74	25.31 ± 0.60	26.83 ± 0.59	27.19 ± 0.86	116.10 ± 7.29	24.42 ± 0.41	29.06 ± 0.76	23.49 ± 0.44	24.56 ± 0.93	25.13 ± 0.85	25.11 ± 0.78	61.09 ± 37.33	25.25 ± 0.88	24.33 ± 0.41	24.51 ± 0.52	35.42 ± 0.64	24.92 ± 0.37	26.15 ± 0.70	25.60	25.64	21.67 ± 0.02	26.75	22.50 ± 0.02	25.96	27.92 ± 0.16	25.93	23.30 ± 0.01	22.83 ± 0.44	**20.54 ± 0.10**	*21.14 ± 0.03*

PEMS06

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	14.29 ± 0.34	14.25 ± 0.15	14.75 ± 0.46	14.91 ± 0.58	78.18 ± 1.49	13.39 ± 0.36	26.65 ± 0.40	13.10 ± 0.23	13.47 ± 0.33	17.23 ± 3.32	13.86 ± 0.26	19.38 ± 4.55	13.53 ± 0.45	13.28 ± 0.36	14.65 ± 0.27	15.70 ± 0.34	14.19 ± 0.45	14.86 ± 0.15	13.18	13.03	12.20 ± 0.01	13.60	12.46 ± 0.01	14.70	17.57 ± 0.43	14.03	12.68 ± 0.01	12.73 ± 0.61	*11.73 ± 0.57*	**11.50 ± 0.14**
	6	14.60 ± 0.30	15.36 ± 0.17	15.33 ± 0.46	15.92 ± 0.53	78.77 ± 1.35	14.24 ± 0.34	27.53 ± 0.65	13.91 ± 0.21	14.33 ± 0.33	17.90 ± 3.21	14.75 ± 0.23	20.83 ± 4.24	14.30 ± 0.46	14.13 ± 0.34	15.20 ± 0.21	16.52 ± 0.33	15.02 ± 0.42	16.16 ± 0.13	15.19	15.11	13.47 ± 0.01	15.79	14.02 ± 0.01	17.33	19.73 ± 0.46	16.01	14.08 ± 0.01	13.09 ± 0.61	**12.15 ± 0.58**	*12.31 ± 0.16*
	12	16.11 ± 0.30	17.62 ± 0.22	16.90 ± 0.44	18.24 ± 0.51	79.36 ± 1.36	15.99 ± 0.31	29.23 ± 1.05	15.54 ± 0.26	16.09 ± 0.35	19.37 ± 2.99	16.67 ± 0.21	23.88 ± 3.70	15.99 ± 0.52	15.88 ± 0.32	16.77 ± 0.20	18.34 ± 0.40	16.83 ± 0.37	18.94 ± 0.11	20.01	19.88	16.06 ± 0.01	20.01	17.21 ± 0.01	22.01	24.25 ± 0.31	20.28	16.91 ± 0.01	13.78 ± 0.56	**12.88 ± 0.58**	*13.77 ± 0.18*
	Avg	14.87 ± 0.31	15.52 ± 0.16	15.52 ± 0.46	16.16 ± 0.54	78.66 ± 1.39	14.37 ± 0.34	27.64 ± 0.65	14.02 ± 0.21	14.46 ± 0.34	18.02 ± 3.19	14.90 ± 0.24	21.10 ± 4.19	14.44 ± 0.46	14.25 ± 0.34	15.39 ± 0.23	16.70 ± 0.35	15.18 ± 0.41	16.40 ± 0.13	15.71	15.59	13.66 ± 0.01	16.08	14.28 ± 0.01	17.67	20.04 ± 0.40	16.33	14.28 ± 0.01	13.13 ± 0.60	**12.18 ± 0.58**	*12.38 ± 0.15*
RMSE	3	22.13 ± 0.48	21.74 ± 0.27	23.99 ± 0.57	22.66 ± 0.70	96.67 ± 2.18	20.57 ± 0.42	36.24 ± 0.70	20.00 ± 0.18	20.57 ± 0.42	27.60 ± 6.52	21.19 ± 0.35	28.19 ± 5.72	20.82 ± 0.45	20.39 ± 0.41	23.12 ± 0.45	23.42 ± 0.45	21.56 ± 0.48	22.68 ± 0.15	20.98	20.74	19.77 ± 0.05	21.65	20.03 ± 0.02	23.69	27.96 ± 0.62	22.24	20.34 ± 0.02	19.15 ± 0.71	**17.89 ± 0.81**	*17.96 ± 0.25*
	6	22.84 ± 0.42	23.84 ± 0.29	25.08 ± 0.56	24.49 ± 0.62	98.00 ± 2.04	22.23 ± 0.39	38.11 ± 1.19	21.54 ± 0.29	22.27 ± 0.46	28.90 ± 6.09	22.98 ± 0.33	30.83 ± 5.09	22.35 ± 0.47	22.06 ± 0.39	24.03 ± 0.42	24.72 ± 0.43	23.16 ± 0.48	25.24 ± 0.14	24.99	24.87	22.65 ± 0.03	25.77	23.64 ± 0.01	28.35	32.03 ± 0.80	25.81	23.30 ± 0.04	19.84 ± 0.70	**18.64 ± 0.79**	*19.55 ± 0.28*
	12	25.62 ± 0.41	27.71 ± 0.34	27.79 ± 0.53	28.42 ± 0.62	100.02 ± 1.93	25.34 ± 0.36	41.47 ± 2.03	24.41 ± 0.45	25.39 ± 0.54	31.36 ± 5.41	26.49 ± 0.34	36.07 ± 4.02	25.35 ± 0.62	25.18 ± 0.37	26.62 ± 0.35	27.54 ± 0.47	26.36 ± 0.43	30.13 ± 0.17	33.98	33.64	28.54 ± 0.08	33.36	30.70 ± 0.03	36.75	44.91 ± 8.74	33.00	29.32 ± 0.08	*21.03 ± 0.61*	**19.85 ± 0.76**	22.20 ± 0.32
	Avg	23.29 ± 0.43	24.04 ± 0.28	25.38 ± 0.56	24.84 ± 0.65	97.96 ± 2.03	22.39 ± 0.39	38.29 ± 1.20	21.68 ± 0.28	22.41 ± 0.47	29.03 ± 6.07	23.19 ± 0.34	31.23 ± 4.99	22.53 ± 0.49	22.21 ± 0.38	24.34 ± 0.40	24.97 ± 0.45	23.39 ± 0.46	25.55 ± 0.15	25.88	25.65	23.12 ± 0.04	26.22	24.21 ± 0.01	28.85	32.90 ± 1.12	26.23	23.75 ± 0.04	19.88 ± 0.68	**18.66 ± 0.79**	*19.62 ± 0.27*
MAPE (%)	3	24.10 ± 0.81	21.76 ± 0.38	23.21 ± 0.43	23.31 ± 0.78	79.39 ± 1.35	21.38 ± 0.30	31.34 ± 0.34	20.89 ± 0.38	21.45 ± 0.51	25.89 ± 5.52	23.18 ± 0.54	34.95 ± 16.21	21.52 ± 0.65	21.25 ± 0.30	21.75 ± 0.39	30.95 ± 1.10	22.76 ± 0.99	22.34 ± 0.37	19.47	19.36	17.99 ± 0.04	20.16	18.36 ± 0.01	19.80	21.95 ± 0.35	19.90	18.89 ± 0.01	19.81 ± 0.84	**18.31 ± 0.45**	*18.52 ± 0.20*
	6	24.33 ± 0.75	23.07 ± 0.47	23.76 ± 0.39	24.42 ± 0.81	79.52 ± 1.43	22.31 ± 0.32	32.02 ± 0.39	21.73 ± 0.47	22.24 ± 0.49	26.69 ± 5.49	23.91 ± 0.56	36.41 ± 15.97	22.36 ± 0.64	22.19 ± 0.32	22.48 ± 0.39	32.23 ± 1.07	23.77 ± 0.94	24.02 ± 0.37	21.56	21.61	19.19 ± 0.04	22.53	19.79 ± 0.01	22.19	24.49 ± 0.32	22.05	20.42 ± 0.01	20.23 ± 0.87	**18.75 ± 0.46**	*19.38 ± 0.26*
	12	25.99 ± 0.73	26.28 ± 0.63	25.75 ± 0.39	27.32 ± 0.86	79.07 ± 1.37	24.56 ± 0.38	33.58 ± 0.89	23.69 ± 0.83	24.14 ± 0.51	28.62 ± 5.53	25.98 ± 0.56	39.24 ± 15.64	24.53 ± 0.73	24.43 ± 0.38	24.45 ± 0.40	35.38 ± 1.06	26.08 ± 0.96	28.21 ± 0.42	27.09	27.27	21.83 ± 0.05	27.51	22.86 ± 0.01	26.87	28.81 ± 0.24	26.63	23.73 ± 0.02	*21.03 ± 0.84*	**19.61 ± 0.46**	21.11 ± 0.39
	Avg	24.66 ± 0.77	23.41 ± 0.46	24.06 ± 0.40	24.76 ± 0.81	79.40 ± 1.33	22.54 ± 0.33	32.16 ± 0.38	21.93 ± 0.52	22.43 ± 0.49	26.88 ± 5.50	24.14 ± 0.58	36.63 ± 16.01	22.60 ± 0.66	22.42 ± 0.33	22.72 ± 0.38	32.58 ± 1.07	24.00 ± 0.96	24.49 ± 0.37	22.24	22.27	19.42 ± 0.04	22.96	20.06 ± 0.01	22.54	24.70 ± 0.29	22.46	20.71 ± 0.01	20.28 ± 0.85	**18.80 ± 0.45**	*19.51 ± 0.27*

PEMS07

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	18.59 ± 0.29	17.59 ± 0.06	19.66 ± 0.44	18.38 ± 1.08	40.22 ± 2.16	16.53 ± 0.25	21.47 ± 0.50	16.22 ± 0.28	16.12 ± 0.30	17.90 ± 1.53	15.62 ± 1.26	17.21 ± 0.25	16.37 ± 0.15	16.24 ± 0.26	18.02 ± 0.25	23.09 ± 0.91	17.17 ± 0.11	17.63 ± 0.04	15.37	15.19	14.38 ± 0.02	15.97	14.96 ± 0.03	18.20	20.28 ± 0.24	17.22	14.98 ± 0.02	14.67 ± 0.34	**13.00 ± 0.06**	*13.77 ± 0.04*
	6	18.93 ± 0.27	19.22 ± 0.05	20.25 ± 0.40	19.21 ± 0.96	40.32 ± 1.94	17.83 ± 0.27	22.58 ± 0.58	17.53 ± 0.32	17.40 ± 0.31	19.09 ± 1.46	16.93 ± 1.25	18.14 ± 0.23	17.64 ± 0.15	17.55 ± 0.27	18.94 ± 0.17	24.77 ± 1.12	18.44 ± 0.08	19.49 ± 0.01	18.26	18.03	16.36 ± 0.03	19.10	17.51 ± 0.02	21.91	23.40 ± 0.16	20.19	17.24 ± 0.01	15.29 ± 0.29	**13.76 ± 0.07**	*15.12 ± 0.06*
	12	21.19 ± 0.27	22.81 ± 0.08	22.49 ± 0.38	21.54 ± 0.96	40.46 ± 2.10	20.40 ± 0.30	24.70 ± 0.74	19.94 ± 0.36	19.79 ± 0.36	21.32 ± 1.38	19.34 ± 1.26	19.99 ± 0.38	20.09 ± 0.22	20.14 ± 0.30	21.55 ± 0.15	28.54 ± 1.56	21.02 ± 0.10	23.49 ± 0.03	25.04	24.47	20.20 ± 0.06	25.20	22.33 ± 0.02	28.49	29.63 ± 0.12	26.07	21.56 ± 0.01	*16.32 ± 0.21*	**14.91 ± 0.10**	17.46 ± 0.09
	Avg	19.39 ± 0.28	19.56 ± 0.05	20.62 ± 0.41	19.51 ± 0.99	40.30 ± 2.05	18.02 ± 0.27	22.72 ± 0.59	17.67 ± 0.31	17.54 ± 0.31	19.23 ± 1.47	17.07 ± 1.26	18.26 ± 0.27	17.81 ± 0.16	17.74 ± 0.27	19.27 ± 0.19	25.17 ± 1.15	18.64 ± 0.09	19.85 ± 0.01	18.98	18.67	16.63 ± 0.04	19.55	17.86 ± 0.02	22.38	23.89 ± 0.16	20.58	17.53 ± 0.01	15.32 ± 0.28	**13.77 ± 0.07**	*15.22 ± 0.06*
RMSE	3	30.83 ± 0.32	29.38 ± 0.14	36.00 ± 0.80	30.15 ± 1.49	52.97 ± 2.13	27.82 ± 0.20	33.25 ± 0.58	27.41 ± 0.29	27.28 ± 0.25	29.96 ± 2.69	26.60 ± 1.78	28.38 ± 0.35	27.62 ± 0.15	27.46 ± 0.21	29.94 ± 0.28	36.09 ± 1.22	28.42 ± 0.19	29.63 ± 0.04	27.86	27.55	26.41 ± 0.05	28.60	27.46 ± 0.05	32.69	36.91 ± 0.74	30.43	27.18 ± 0.07	25.56 ± 0.32	**23.51 ± 0.08**	*24.67 ± 0.05*
	6	31.89 ± 0.28	32.54 ± 0.12	37.21 ± 0.75	31.78 ± 1.30	54.08 ± 1.79	30.46 ± 0.22	35.59 ± 0.67	29.94 ± 0.33	29.84 ± 0.28	32.30 ± 2.45	29.14 ± 1.82	30.27 ± 0.30	30.13 ± 0.17	30.10 ± 0.22	31.73 ± 0.17	38.85 ± 1.49	30.90 ± 0.16	33.33 ± 0.06	33.35	32.91	30.53 ± 0.10	34.44	32.80 ± 0.03	39.61	42.56 ± 0.17	35.69	31.79 ± 0.06	*26.92 ± 0.27*	**25.09 ± 0.08**	27.21 ± 0.09
	12	36.27 ± 0.30	39.02 ± 0.14	41.01 ± 0.68	35.84 ± 1.30	55.87 ± 1.97	35.29 ± 0.28	39.86 ± 0.86	34.34 ± 0.39	34.29 ± 0.38	36.50 ± 2.15	33.55 ± 1.93	33.61 ± 0.60	34.67 ± 0.25	34.96 ± 0.28	36.28 ± 0.20	45.39 ± 2.19	35.56 ± 0.17	40.75 ± 0.12	46.69	45.31	38.69 ± 0.25	46.03	42.92 ± 0.05	52.40	59.72 ± 6.83	46.50	40.60 ± 0.03	*28.97 ± 0.18*	**27.33 ± 0.11**	31.28 ± 0.14
	Avg	32.62 ± 0.30	33.05 ± 0.13	37.75 ± 0.75	32.22 ± 1.35	54.10 ± 1.93	30.73 ± 0.23	35.82 ± 0.68	30.13 ± 0.32	30.04 ± 0.29	32.52 ± 2.46	29.33 ± 1.84	30.41 ± 0.37	30.37 ± 0.18	30.37 ± 0.23	32.22 ± 0.21	39.57 ± 1.56	31.19 ± 0.17	33.90 ± 0.07	34.86	34.18	31.15 ± 0.13	35.34	33.57 ± 0.03	40.57	44.62 ± 1.28	36.42	32.42 ± 0.05	*26.93 ± 0.26*	**25.05 ± 0.08**	27.30 ± 0.08
MAPE (%)	3	30.48 ± 1.24	27.98 ± 0.33	31.59 ± 1.03	30.73 ± 2.59	50.26 ± 2.43	26.51 ± 1.30	34.13 ± 1.18	25.57 ± 1.32	25.51 ± 2.27	26.75 ± 1.59	24.19 ± 3.12	27.34 ± 0.96	25.71 ± 0.67	26.27 ± 1.27	32.44 ± 0.94	55.33 ± 2.34	29.41 ± 1.13	27.44 ± 0.37	19.45	19.33	18.12 ± 0.03	20.30	18.88 ± 0.03	20.43	21.13 ± 0.06	20.34	18.90 ± 0.03	20.97 ± 0.86	**18.51 ± 0.27**	*19.46 ± 0.37*
	6	30.62 ± 1.27	29.47 ± 0.24	31.86 ± 1.01	31.46 ± 2.35	49.24 ± 2.48	27.39 ± 1.34	34.81 ± 1.48	26.44 ± 1.29	26.43 ± 2.20	27.64 ± 1.71	25.38 ± 3.00	28.43 ± 0.54	26.65 ± 0.63	27.16 ± 1.32	33.59 ± 0.86	58.00 ± 2.92	30.46 ± 1.12	29.34 ± 0.46	22.00	21.88	19.74 ± 0.04	23.14	20.91 ± 0.02	23.33	24.52 ± 0.11	23.12	20.85 ± 0.03	21.31 ± 0.65	**18.99 ± 0.18**	*20.67 ± 0.39*
	12	32.67 ± 1.31	33.46 ± 0.27	34.54 ± 1.08	33.85 ± 2.10	47.91 ± 3.13	29.62 ± 1.46	36.82 ± 2.23	28.63 ± 1.18	28.65 ± 2.11	29.78 ± 1.91	28.11 ± 3.10	31.10 ± 1.22	28.92 ± 0.73	29.40 ± 1.43	37.31 ± 0.74	64.27 ± 4.24	33.08 ± 1.16	34.30 ± 0.78	28.62	28.40	23.16 ± 0.04	29.16	24.96 ± 0.02	28.97	29.96 ± 0.09	28.69	24.87 ± 0.05	*22.17 ± 0.41*	**20.02 ± 0.14**	23.20 ± 0.40
	Avg	31.09 ± 1.27	29.95 ± 0.26	32.43 ± 1.02	31.81 ± 2.36	49.24 ± 2.54	27.64 ± 1.35	35.08 ± 1.56	26.68 ± 1.27	26.67 ± 2.19	27.85 ± 1.72	25.66 ± 3.04	28.73 ± 0.39	26.88 ± 0.66	27.40 ± 1.33	34.13 ± 0.84	58.70 ± 3.03	30.76 ± 1.13	29.94 ± 0.50	22.81	22.66	20.04 ± 0.03	23.68	21.24 ± 0.02	23.73	24.80 ± 0.09	23.57	21.17 ± 0.03	21.39 ± 0.65	**19.07 ± 0.20**	*20.90 ± 0.39*

PEMS08

Metric	**Horizon**	**DCRNN**	**ASTGNN**	**TGCN**	**GWN**	**Pretrain**	**Retrain**	**OL-NN**	**OL-AN**	**TrafStm**	**PECPM**	**STKEC**	**EAC**	**STRAP**	**ST-TTC**	**STID**	**STNorm**	**iTrans**	**DLinear**	**Chronos2-U**	**Chronos2-M**	**Chronos2-FT-M**	**TimesFM**	**TimesFM-FT**	**MoE-ZS**	**MoE-FT**	**Moirai2-ZS**	**Moirai2-FT**	**STAEFormer**	**STAE-Ours**	**A2TTA**
MAE	3	14.86 ± 0.14	13.63 ± 0.05	15.01 ± 0.19	14.62 ± 0.19	52.44 ± 2.51	13.13 ± 0.06	16.14 ± 0.04	12.76 ± 0.09	12.85 ± 0.12	14.58 ± 0.83	13.32 ± 0.44	13.86 ± 0.28	12.98 ± 0.04	12.95 ± 0.06	14.23 ± 0.08	17.19 ± 0.35	13.42 ± 0.09	13.79 ± 0.05	13.40	13.26	12.46 ± 0.01	13.94	12.77 ± 0.02	15.65	17.49 ± 0.13	14.90	13.02 ± 0.02	11.99 ± 0.04	**11.20 ± 0.02**	*11.87 ± 0.03*
	6	15.17 ± 0.13	14.97 ± 0.05	15.60 ± 0.16	15.39 ± 0.16	52.24 ± 2.61	14.18 ± 0.07	17.16 ± 0.06	13.80 ± 0.10	13.91 ± 0.12	15.47 ± 0.78	14.31 ± 0.33	14.66 ± 0.28	14.02 ± 0.03	14.00 ± 0.06	14.83 ± 0.06	18.07 ± 0.30	14.44 ± 0.08	15.28 ± 0.04	15.75	15.62	14.00 ± 0.02	16.49	14.62 ± 0.01	18.77	19.98 ± 0.13	17.30	14.75 ± 0.02	*12.47 ± 0.03*	**11.74 ± 0.03**	12.88 ± 0.04
	12	16.88 ± 0.13	17.84 ± 0.05	17.43 ± 0.16	17.61 ± 0.20	52.07 ± 3.05	16.27 ± 0.09	19.13 ± 0.11	15.75 ± 0.11	15.90 ± 0.17	17.19 ± 0.67	16.32 ± 0.36	16.25 ± 0.27	16.00 ± 0.03	16.10 ± 0.08	16.83 ± 0.06	20.27 ± 0.37	16.48 ± 0.09	18.53 ± 0.04	21.43	21.09	17.01 ± 0.04	21.51	18.18 ± 0.01	24.43	25.47 ± 0.26	22.15	18.10 ± 0.03	*13.32 ± 0.03*	**12.63 ± 0.04**	14.67 ± 0.06
	Avg	15.50 ± 0.13	15.22 ± 0.05	15.86 ± 0.17	15.69 ± 0.16	52.25 ± 2.71	14.34 ± 0.07	17.30 ± 0.05	13.92 ± 0.10	14.04 ± 0.13	15.58 ± 0.77	14.47 ± 0.38	14.76 ± 0.28	14.15 ± 0.03	14.16 ± 0.07	15.12 ± 0.07	18.33 ± 0.31	14.60 ± 0.08	15.58 ± 0.04	16.39	16.19	14.22 ± 0.02	16.87	14.90 ± 0.01	19.21	20.52 ± 0.16	17.64	14.99 ± 0.02	*12.52 ± 0.03*	**11.76 ± 0.03**	12.97 ± 0.04
RMSE	3	23.73 ± 0.20	22.36 ± 0.08	26.25 ± 0.43	23.28 ± 0.29	61.63 ± 2.94	21.49 ± 0.07	24.62 ± 0.06	21.01 ± 0.12	21.09 ± 0.13	23.27 ± 1.20	21.64 ± 0.51	21.99 ± 0.33	21.29 ± 0.07	21.26 ± 0.05	24.03 ± 0.21	26.14 ± 0.40	21.79 ± 0.12	22.74 ± 0.04	22.55	22.30	21.40 ± 0.03	23.20	21.87 ± 0.03	26.33	29.66 ± 0.56	24.72	22.10 ± 0.05	*19.83 ± 0.09*	**18.81 ± 0.05**	19.93 ± 0.04
	6	24.70 ± 0.18	24.88 ± 0.10	27.48 ± 0.38	24.83 ± 0.26	62.11 ± 3.02	23.63 ± 0.09	26.65 ± 0.12	23.05 ± 0.15	23.19 ± 0.16	25.08 ± 1.05	23.66 ± 0.40	23.65 ± 0.34	23.35 ± 0.04	23.41 ± 0.08	25.24 ± 0.16	27.58 ± 0.31	23.77 ± 0.13	25.65 ± 0.03	26.85	26.57	24.61 ± 0.06	27.72	25.66 ± 0.02	31.97	33.94 ± 0.36	28.76	25.56 ± 0.05	*20.82 ± 0.08*	**19.91 ± 0.06**	21.88 ± 0.07
	12	28.02 ± 0.18	29.80 ± 0.13	30.67 ± 0.37	28.68 ± 0.40	63.24 ± 3.50	27.52 ± 0.13	30.31 ± 0.23	26.53 ± 0.20	26.81 ± 0.29	28.33 ± 0.80	27.41 ± 0.62	26.49 ± 0.45	27.00 ± 0.05	27.31 ± 0.12	28.72 ± 0.14	31.18 ± 0.47	27.42 ± 0.17	31.37 ± 0.05	37.17	36.33	30.88 ± 0.15	36.53	32.74 ± 0.02	41.51	47.49 ± 1.38	37.04	31.97 ± 0.08	*22.44 ± 0.10*	**21.57 ± 0.09**	25.03 ± 0.11
	Avg	25.20 ± 0.19	25.23 ± 0.10	27.85 ± 0.40	25.27 ± 0.29	62.21 ± 3.13	23.85 ± 0.09	26.86 ± 0.13	23.20 ± 0.15	23.35 ± 0.18	25.25 ± 1.04	23.89 ± 0.50	23.76 ± 0.37	23.54 ± 0.04	23.63 ± 0.08	25.69 ± 0.17	28.00 ± 0.33	23.99 ± 0.13	26.07 ± 0.04	28.01	27.58	25.08 ± 0.08	28.38	26.18 ± 0.02	32.55	35.46 ± 0.39	29.31	25.97 ± 0.06	*20.88 ± 0.09*	**19.91 ± 0.06**	21.96 ± 0.07
MAPE (%)	3	22.82 ± 0.42	19.07 ± 0.41	21.25 ± 0.69	20.76 ± 0.73	57.52 ± 1.13	19.25 ± 0.27	22.79 ± 0.31	18.05 ± 0.22	19.62 ± 1.41	21.16 ± 1.57	20.30 ± 1.59	26.19 ± 1.99	18.63 ± 0.35	19.13 ± 0.27	19.38 ± 0.76	41.36 ± 1.88	20.82 ± 1.06	18.96 ± 0.43	17.10	17.01	15.85 ± 0.02	17.93	16.36 ± 0.01	18.02	18.90 ± 0.04	17.86	16.69 ± 0.04	17.16 ± 0.26	**16.02 ± 0.07**	*16.31 ± 0.10*
	6	22.94 ± 0.42	20.47 ± 0.38	21.71 ± 0.71	21.51 ± 0.73	56.94 ± 1.54	20.33 ± 0.24	23.83 ± 0.33	19.02 ± 0.21	20.70 ± 1.34	22.07 ± 1.65	21.30 ± 1.42	26.35 ± 1.97	19.69 ± 0.40	20.20 ± 0.23	19.98 ± 0.75	42.87 ± 1.97	21.81 ± 1.00	20.66 ± 0.40	19.36	19.33	17.22 ± 0.03	20.46	17.97 ± 0.00	20.63	21.71 ± 0.05	20.31	18.33 ± 0.04	17.53 ± 0.25	**16.42 ± 0.07**	*17.28 ± 0.09*
	12	24.58 ± 0.45	23.92 ± 0.38	24.00 ± 0.73	24.07 ± 0.76	56.54 ± 2.39	22.74 ± 0.27	26.26 ± 0.49	21.37 ± 0.32	23.18 ± 1.24	24.14 ± 1.73	23.62 ± 0.94	27.80 ± 2.10	22.19 ± 0.54	22.61 ± 0.27	22.15 ± 0.79	46.66 ± 2.26	24.16 ± 0.88	24.95 ± 0.43	25.40	25.32	20.17 ± 0.04	25.89	21.27 ± 0.00	25.69	26.68 ± 0.07	25.19	21.82 ± 0.04	*18.49 ± 0.24*	**17.33 ± 0.09**	19.33 ± 0.10
	Avg	23.31 ± 0.41	20.85 ± 0.39	22.13 ± 0.71	21.90 ± 0.74	57.09 ± 1.58	20.58 ± 0.25	24.09 ± 0.35	19.28 ± 0.22	20.96 ± 1.33	22.26 ± 1.64	21.55 ± 1.34	26.61 ± 1.73	19.96 ± 0.41	20.45 ± 0.25	20.33 ± 0.76	43.33 ± 2.01	22.07 ± 0.99	21.16 ± 0.42	20.13	20.06	17.49 ± 0.03	20.96	18.26 ± 0.00	21.00	22.04 ± 0.05	20.70	18.63 ± 0.04	17.66 ± 0.23	**16.51 ± 0.07**	*17.47 ± 0.10*

PEMS10

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	11.81 ± 0.19	11.46 ± 0.05	11.82 ± 0.09	11.87 ± 0.11	81.21 ± 3.23	11.09 ± 0.04	11.49 ± 0.20	10.86 ± 0.08	11.14 ± 0.06	11.90 ± 1.24	11.15 ± 0.08	12.84 ± 1.70	11.05 ± 0.08	11.01 ± 0.04	11.85 ± 0.09	13.13 ± 0.29	11.13 ± 0.07	11.73 ± 0.04	11.64	11.46	10.79 ± 0.01	11.82	11.04 ± 0.01	13.03	14.44 ± 0.12	12.56	11.24 ± 0.00	*10.09 ± 0.03*	**9.59 ± 0.03**	10.23 ± 0.03
	6	12.06 ± 0.14	12.32 ± 0.05	12.29 ± 0.08	12.38 ± 0.11	81.43 ± 3.27	11.74 ± 0.04	12.23 ± 0.23	11.50 ± 0.10	11.82 ± 0.08	12.51 ± 1.18	11.81 ± 0.07	13.48 ± 1.73	11.68 ± 0.09	11.65 ± 0.04	12.36 ± 0.07	13.62 ± 0.32	11.74 ± 0.08	12.69 ± 0.04	13.06	12.93	11.74 ± 0.01	13.38	12.10 ± 0.01	15.01	16.15 ± 0.13	13.93	12.30 ± 0.00	*10.32 ± 0.03*	**9.84 ± 0.03**	10.81 ± 0.04
	12	13.19 ± 0.15	14.05 ± 0.05	13.48 ± 0.08	13.70 ± 0.14	81.53 ± 3.26	13.06 ± 0.06	13.70 ± 0.29	12.76 ± 0.12	13.18 ± 0.13	13.72 ± 1.04	13.14 ± 0.08	14.87 ± 1.91	12.97 ± 0.10	12.97 ± 0.05	13.73 ± 0.09	14.81 ± 0.35	12.95 ± 0.07	14.67 ± 0.04	16.56	16.37	13.51 ± 0.01	16.49	14.17 ± 0.01	18.64	19.70 ± 0.22	16.97	14.30 ± 0.01	*10.73 ± 0.04*	**10.26 ± 0.03**	11.91 ± 0.06
	Avg	12.26 ± 0.16	12.45 ± 0.05	12.43 ± 0.08	12.54 ± 0.11	81.39 ± 3.16	11.85 ± 0.04	12.35 ± 0.23	11.60 ± 0.09	11.93 ± 0.08	12.61 ± 1.16	11.92 ± 0.07	13.60 ± 1.76	11.79 ± 0.09	11.76 ± 0.04	12.53 ± 0.08	13.75 ± 0.32	11.83 ± 0.07	12.86 ± 0.04	13.47	13.30	11.85 ± 0.01	13.63	12.27 ± 0.01	15.31	16.45 ± 0.14	14.18	12.43 ± 0.01	*10.34 ± 0.03*	**9.86 ± 0.03**	10.88 ± 0.04
RMSE	3	19.17 ± 0.29	19.03 ± 0.09	20.39 ± 0.16	19.21 ± 0.12	83.86 ± 3.27	18.21 ± 0.02	18.96 ± 0.38	17.84 ± 0.14	18.31 ± 0.08	20.00 ± 2.91	18.35 ± 0.11	19.71 ± 1.41	18.17 ± 0.10	18.08 ± 0.02	19.33 ± 0.09	20.78 ± 0.37	18.33 ± 0.09	19.35 ± 0.05	19.62	19.29	18.45 ± 0.03	19.75	18.67 ± 0.01	21.83	23.87 ± 0.10	20.99	19.00 ± 0.02	*16.69 ± 0.07*	**16.07 ± 0.04**	17.08 ± 0.05
	6	19.79 ± 0.24	20.59 ± 0.06	21.24 ± 0.14	20.20 ± 0.12	84.50 ± 3.37	19.49 ± 0.04	20.39 ± 0.45	19.06 ± 0.18	19.58 ± 0.14	21.13 ± 2.71	19.63 ± 0.10	20.94 ± 1.41	19.40 ± 0.10	19.35 ± 0.04	20.30 ± 0.07	21.60 ± 0.41	19.49 ± 0.10	21.11 ± 0.05	22.12	21.87	20.39 ± 0.04	22.48	20.80 ± 0.01	25.46	28.17 ± 2.25	23.25	21.08 ± 0.02	*17.17 ± 0.05*	**16.57 ± 0.03**	18.16 ± 0.08
	12	21.96 ± 0.25	23.58 ± 0.05	23.35 ± 0.13	22.52 ± 0.20	85.41 ± 3.41	21.94 ± 0.07	23.10 ± 0.55	21.39 ± 0.21	22.03 ± 0.24	23.31 ± 2.32	22.08 ± 0.12	23.29 ± 1.57	21.80 ± 0.14	21.80 ± 0.07	22.59 ± 0.11	23.56 ± 0.44	21.66 ± 0.11	24.54 ± 0.05	28.44	27.96	24.06 ± 0.05	27.93	24.87 ± 0.01	32.24	33.91 ± 1.29	28.30	24.93 ± 0.04	*17.96 ± 0.05*	**17.36 ± 0.03**	20.18 ± 0.11
	Avg	20.13 ± 0.26	20.80 ± 0.06	21.49 ± 0.14	20.45 ± 0.13	84.52 ± 3.24	19.66 ± 0.03	20.58 ± 0.45	19.22 ± 0.17	19.77 ± 0.14	21.29 ± 2.68	19.80 ± 0.11	21.08 ± 1.43	19.58 ± 0.10	19.53 ± 0.04	20.54 ± 0.09	21.81 ± 0.40	19.63 ± 0.10	21.36 ± 0.05	22.89	22.54	20.64 ± 0.04	22.92	21.11 ± 0.01	25.99	28.09 ± 1.21	23.64	21.32 ± 0.03	*17.20 ± 0.05*	**16.58 ± 0.03**	18.28 ± 0.08
MAPE (%)	3	30.74 ± 0.34	28.38 ± 0.32	29.73 ± 0.29	29.49 ± 0.74	756.76 ± 29.47	29.11 ± 0.29	30.04 ± 0.94	28.70 ± 0.37	30.52 ± 0.75	29.20 ± 1.71	29.69 ± 0.29	58.30 ± 38.83	29.13 ± 0.55	28.95 ± 0.28	31.70 ± 0.31	40.22 ± 1.74	29.02 ± 0.61	28.77 ± 0.41	28.04	27.81	26.36 ± 0.07	28.80	27.03 ± 0.03	28.16	29.19 ± 0.16	28.44	27.36 ± 0.03	27.39 ± 0.42	**25.98 ± 0.03**	*26.87 ± 0.08*
	6	31.29 ± 0.33	29.84 ± 0.32	30.52 ± 0.23	30.31 ± 0.69	755.22 ± 29.03	30.00 ± 0.23	31.38 ± 0.98	29.47 ± 0.34	31.73 ± 0.67	30.11 ± 1.70	30.78 ± 0.50	58.55 ± 40.02	30.04 ± 0.59	29.84 ± 0.22	32.71 ± 0.23	41.21 ± 1.76	30.02 ± 0.71	30.57 ± 0.38	29.87	29.81	27.60 ± 0.07	30.98	28.35 ± 0.03	30.54	31.69 ± 0.10	30.28	28.91 ± 0.03	27.74 ± 0.45	**26.34 ± 0.05**	*27.55 ± 0.07*
	12	33.45 ± 0.41	33.35 ± 0.34	32.93 ± 0.18	32.78 ± 0.66	752.24 ± 26.19	32.42 ± 0.18	34.68 ± 1.59	31.77 ± 0.42	35.13 ± 0.79	32.53 ± 1.65	33.28 ± 0.51	62.45 ± 42.54	32.43 ± 0.61	32.25 ± 0.17	35.68 ± 0.25	43.96 ± 1.88	32.40 ± 0.94	34.98 ± 0.32	35.06	35.14	30.28 ± 0.06	35.86	31.18 ± 0.03	35.72	36.23 ± 0.13	34.51	32.29 ± 0.03	*28.55 ± 0.44*	**27.08 ± 0.07**	29.19 ± 0.10
	Avg	31.65 ± 0.35	30.22 ± 0.32	30.86 ± 0.22	30.65 ± 0.69	755.06 ± 28.01	30.30 ± 0.22	31.79 ± 1.11	29.80 ± 0.34	32.18 ± 0.70	30.41 ± 1.69	31.06 ± 0.42	59.52 ± 40.27	30.34 ± 0.58	30.14 ± 0.22	33.11 ± 0.25	41.57 ± 1.79	30.28 ± 0.74	31.07 ± 0.38	30.60	30.51	27.84 ± 0.07	31.47	28.62 ± 0.03	30.97	32.02 ± 0.11	30.73	29.22 ± 0.02	27.83 ± 0.45	**26.41 ± 0.04**	*27.73 ± 0.07*

PEMS11

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	18.69 ± 0.36	17.88 ± 0.20	20.19 ± 0.31	18.90 ± 0.41	83.38 ± 3.81	17.03 ± 0.13	19.74 ± 1.56	16.62 ± 0.13	16.93 ± 0.27	24.48 ± 2.64	17.24 ± 0.28	23.71 ± 3.81	16.91 ± 0.12	16.76 ± 0.12	18.57 ± 0.32	20.18 ± 0.43	17.54 ± 0.29	18.32 ± 0.15	15.94	15.80	14.70 ± 0.01	16.84	15.25 ± 0.02	18.69	20.45 ± 0.14	17.40	15.09 ± 0.00	16.33 ± 0.16	**13.40 ± 0.06**	*13.97 ± 0.07*
	6	19.19 ± 0.28	19.63 ± 0.17	20.93 ± 0.31	19.91 ± 0.39	85.10 ± 3.52	18.50 ± 0.12	21.22 ± 1.61	18.07 ± 0.12	18.43 ± 0.34	25.70 ± 2.54	18.77 ± 0.21	25.31 ± 4.21	18.36 ± 0.15	18.26 ± 0.12	19.61 ± 0.32	21.46 ± 0.36	18.98 ± 0.29	20.20 ± 0.11	19.31	19.21	16.84 ± 0.02	20.39	17.88 ± 0.02	22.74	24.00 ± 0.10	20.82	17.34 ± 0.01	16.94 ± 0.14	**14.32 ± 0.07**	*15.44 ± 0.08*
	12	21.83 ± 0.24	23.48 ± 0.12	23.55 ± 0.32	22.77 ± 0.55	88.77 ± 3.23	21.61 ± 0.15	24.45 ± 1.70	20.92 ± 0.13	21.44 ± 0.47	28.19 ± 2.36	21.86 ± 0.16	29.22 ± 4.63	21.32 ± 0.24	21.38 ± 0.15	22.64 ± 0.34	24.40 ± 0.67	21.99 ± 0.27	24.43 ± 0.09	27.46	27.09	21.20 ± 0.04	27.29	23.15 ± 0.03	29.56	30.42 ± 0.16	27.38	21.94 ± 0.03	18.28 ± 0.12	**15.96 ± 0.09**	*18.12 ± 0.10*
	Avg	19.69 ± 0.30	20.00 ± 0.16	21.35 ± 0.31	20.29 ± 0.44	85.43 ± 3.52	18.77 ± 0.13	21.54 ± 1.59	18.29 ± 0.13	18.67 ± 0.35	25.89 ± 2.53	19.02 ± 0.23	25.76 ± 4.14	18.60 ± 0.15	18.52 ± 0.13	20.02 ± 0.32	21.77 ± 0.34	19.24 ± 0.28	20.62 ± 0.12	20.23	20.04	17.19 ± 0.02	20.91	18.32 ± 0.02	23.16	24.37 ± 0.09	21.24	17.71 ± 0.01	17.05 ± 0.12	**14.40 ± 0.07**	*15.60 ± 0.08*
RMSE	3	31.04 ± 0.45	29.94 ± 0.35	37.69 ± 0.55	31.36 ± 0.74	112.94 ± 3.71	28.37 ± 0.17	32.65 ± 2.94	27.81 ± 0.14	28.11 ± 0.36	42.15 ± 5.33	28.52 ± 0.27	37.06 ± 5.06	28.19 ± 0.16	27.97 ± 0.16	30.85 ± 0.38	32.56 ± 0.56	29.15 ± 0.31	30.60 ± 0.15	28.59	28.35	26.78 ± 0.04	30.08	27.62 ± 0.04	33.70	36.68 ± 0.41	31.28	27.08 ± 0.01	27.05 ± 0.36	**23.33 ± 0.09**	*24.48 ± 0.12*
	6	32.77 ± 0.30	33.71 ± 0.26	39.44 ± 0.53	33.73 ± 0.69	116.32 ± 3.41	31.76 ± 0.14	35.91 ± 2.87	31.05 ± 0.11	31.50 ± 0.44	44.47 ± 4.94	31.98 ± 0.20	40.37 ± 5.70	31.47 ± 0.20	31.38 ± 0.13	33.20 ± 0.37	35.12 ± 0.52	32.37 ± 0.31	34.80 ± 0.12	35.64	35.45	31.71 ± 0.06	37.19	33.61 ± 0.08	41.63	43.90 ± 0.27	37.82	32.06 ± 0.04	28.65 ± 0.25	**25.63 ± 0.10**	*27.84 ± 0.12*
	12	38.65 ± 0.27	41.33 ± 0.17	44.47 ± 0.51	39.49 ± 1.05	123.41 ± 3.10	38.27 ± 0.17	42.48 ± 2.88	36.98 ± 0.16	37.79 ± 0.69	49.10 ± 4.27	38.39 ± 0.19	47.69 ± 6.19	37.68 ± 0.32	37.94 ± 0.17	39.19 ± 0.37	40.62 ± 1.15	38.55 ± 0.30	43.44 ± 0.13	52.40	51.41	41.40 ± 0.19	50.75	44.93 ± 0.10	54.25	56.11 ± 0.11	50.35	41.93 ± 0.10	*31.70 ± 0.19*	**29.38 ± 0.12**	33.49 ± 0.16
	Avg	33.66 ± 0.34	34.32 ± 0.27	40.11 ± 0.53	34.35 ± 0.80	116.95 ± 3.38	32.21 ± 0.15	36.45 ± 2.86	31.40 ± 0.13	31.90 ± 0.47	44.81 ± 4.91	32.38 ± 0.23	41.08 ± 5.52	31.88 ± 0.20	31.83 ± 0.15	33.89 ± 0.36	35.62 ± 0.53	32.79 ± 0.31	35.52 ± 0.13	37.49	37.05	32.41 ± 0.08	38.14	34.43 ± 0.07	42.24	44.37 ± 0.19	38.56	32.78 ± 0.04	28.82 ± 0.20	**25.74 ± 0.10**	*28.06 ± 0.13*
MAPE (%)	3	30.78 ± 0.93	24.39 ± 0.34	29.45 ± 0.64	26.32 ± 0.78	77.32 ± 15.78	26.11 ± 0.80	28.92 ± 0.90	23.82 ± 0.70	26.53 ± 1.24	35.21 ± 6.88	28.98 ± 1.96	45.25 ± 14.22	27.10 ± 0.83	25.97 ± 0.79	30.64 ± 1.81	45.43 ± 2.56	29.58 ± 2.86	24.98 ± 0.73	21.78	21.66	20.03 ± 0.04	22.80	20.81 ± 0.03	22.44	23.02 ± 0.01	22.43	20.95 ± 0.01	23.94 ± 0.63	**20.75 ± 0.22**	*22.48 ± 0.39*
	6	30.74 ± 0.87	26.12 ± 0.37	29.65 ± 0.33	27.14 ± 0.80	80.97 ± 18.12	27.25 ± 0.78	30.36 ± 1.20	25.01 ± 0.79	27.76 ± 1.43	36.53 ± 6.71	30.24 ± 1.63	46.71 ± 12.65	28.41 ± 1.01	27.11 ± 0.78	31.58 ± 1.76	47.41 ± 2.67	30.58 ± 2.79	26.94 ± 0.61	24.49	24.49	21.62 ± 0.04	25.85	22.74 ± 0.04	25.41	26.42 ± 0.04	25.28	22.88 ± 0.02	24.36 ± 0.69	**21.31 ± 0.25**	*23.52 ± 0.39*
	12	32.84 ± 0.78	30.75 ± 0.53	32.76 ± 0.27	29.87 ± 0.87	89.84 ± 23.97	30.48 ± 0.87	34.14 ± 1.85	28.09 ± 0.80	31.40 ± 2.51	39.89 ± 6.43	33.90 ± 1.15	51.27 ± 11.76	31.92 ± 1.30	30.33 ± 0.87	35.10 ± 1.52	52.32 ± 3.03	33.38 ± 2.70	32.38 ± 0.59	31.67	31.70	25.05 ± 0.03	32.22	26.79 ± 0.04	30.99	31.71 ± 0.03	31.06	26.96 ± 0.05	*25.71 ± 0.78*	**22.59 ± 0.30**	25.80 ± 0.38
	Avg	31.28 ± 0.86	26.69 ± 0.39	30.37 ± 0.32	27.55 ± 0.81	82.12 ± 18.72	27.67 ± 0.81	30.83 ± 1.25	25.39 ± 0.75	28.29 ± 1.63	36.90 ± 6.66	30.78 ± 1.62	47.34 ± 12.72	28.85 ± 1.00	27.53 ± 0.80	32.17 ± 1.75	48.00 ± 2.68	30.95 ± 2.77	27.66 ± 0.61	25.41	25.36	21.93 ± 0.04	26.41	23.11 ± 0.03	25.78	26.67 ± 0.03	25.78	23.23 ± 0.03	24.54 ± 0.69	**21.44 ± 0.25**	*23.73 ± 0.38*

PEMS12

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	15.43 ± 0.22	14.23 ± 0.12	16.26 ± 0.09	15.16 ± 0.28	15.17 ± 0.18	13.59 ± 0.05	16.25 ± 0.47	13.24 ± 0.05	13.42 ± 0.06	14.97 ± 1.63	13.42 ± 0.10	13.96 ± 0.25	13.59 ± 0.15	13.41 ± 0.05	15.00 ± 0.09	17.97 ± 0.53	13.92 ± 0.13	14.31 ± 0.02	13.68	13.51	12.73 ± 0.00	14.31	13.24 ± 0.03	15.91	17.32 ± 0.62	14.88	13.21 ± 0.01	12.39 ± 0.11	**11.37 ± 0.04**	*12.11 ± 0.02*
	6	15.76 ± 0.16	15.67 ± 0.09	16.80 ± 0.08	16.11 ± 0.25	16.28 ± 0.20	14.75 ± 0.05	17.78 ± 0.56	14.40 ± 0.06	14.60 ± 0.06	16.02 ± 1.50	14.57 ± 0.08	14.84 ± 0.28	14.72 ± 0.14	14.57 ± 0.05	15.76 ± 0.07	19.17 ± 0.41	15.06 ± 0.10	15.90 ± 0.02	16.23	16.05	14.41 ± 0.01	17.06	15.37 ± 0.03	19.07	20.17 ± 0.58	17.51	15.06 ± 0.01	*12.87 ± 0.10*	**11.94 ± 0.05**	13.20 ± 0.01
	12	17.77 ± 0.14	18.87 ± 0.10	18.83 ± 0.06	18.46 ± 0.20	18.34 ± 0.30	17.02 ± 0.06	20.91 ± 0.77	16.58 ± 0.09	16.81 ± 0.13	18.11 ± 1.30	16.75 ± 0.10	16.63 ± 0.44	16.95 ± 0.14	16.87 ± 0.07	18.07 ± 0.06	22.03 ± 0.30	17.48 ± 0.08	19.37 ± 0.03	22.36	21.94	17.73 ± 0.04	22.48	19.51 ± 0.03	24.73	25.32 ± 0.63	22.80	18.67 ± 0.01	*13.70 ± 0.12*	**12.81 ± 0.07**	15.09 ± 0.02
	Avg	16.16 ± 0.17	15.98 ± 0.10	17.13 ± 0.08	16.38 ± 0.23	16.41 ± 0.21	14.91 ± 0.05	18.04 ± 0.57	14.54 ± 0.06	14.74 ± 0.07	16.18 ± 1.49	14.71 ± 0.08	14.98 ± 0.30	14.88 ± 0.14	14.74 ± 0.05	16.07 ± 0.07	19.49 ± 0.39	15.27 ± 0.11	16.23 ± 0.02	16.92	16.67	14.66 ± 0.01	17.48	15.69 ± 0.02	19.49	20.49 ± 0.59	17.90	15.32 ± 0.01	*12.91 ± 0.11*	**11.95 ± 0.05**	13.28 ± 0.01
RMSE	3	26.80 ± 0.31	25.07 ± 0.18	31.84 ± 0.21	26.13 ± 0.45	25.43 ± 0.16	23.87 ± 0.08	28.37 ± 0.92	23.41 ± 0.03	23.68 ± 0.09	26.16 ± 3.54	23.69 ± 0.15	24.14 ± 0.31	23.87 ± 0.18	23.58 ± 0.05	26.16 ± 0.22	29.50 ± 0.83	24.26 ± 0.18	25.36 ± 0.03	24.78	24.46	23.44 ± 0.02	25.67	24.31 ± 0.05	29.08	32.10 ± 1.13	26.66	24.10 ± 0.03	22.03 ± 0.12	**20.53 ± 0.06**	*21.80 ± 0.03*
	6	27.76 ± 0.23	28.00 ± 0.16	32.92 ± 0.19	28.03 ± 0.39	27.77 ± 0.22	26.29 ± 0.06	31.54 ± 1.11	25.78 ± 0.06	26.12 ± 0.09	28.34 ± 3.19	26.08 ± 0.12	26.01 ± 0.36	26.23 ± 0.17	26.03 ± 0.04	27.64 ± 0.19	31.61 ± 0.57	26.59 ± 0.14	28.70 ± 0.05	29.93	29.57	27.08 ± 0.04	31.07	29.01 ± 0.05	35.26	37.46 ± 1.02	31.59	28.07 ± 0.03	*23.09 ± 0.11*	**21.73 ± 0.07**	23.97 ± 0.02
	12	31.86 ± 0.20	34.05 ± 0.23	36.42 ± 0.17	32.38 ± 0.27	31.84 ± 0.41	30.73 ± 0.07	37.79 ± 1.53	29.93 ± 0.15	30.40 ± 0.22	32.39 ± 2.68	30.26 ± 0.19	29.33 ± 0.71	30.58 ± 0.21	30.50 ± 0.08	31.73 ± 0.20	36.73 ± 0.45	31.16 ± 0.13	35.51 ± 0.08	42.62	41.53	34.49 ± 0.19	41.80	38.08 ± 0.07	46.15	47.62 ± 0.88	41.70	35.73 ± 0.01	*24.77 ± 0.15*	**23.45 ± 0.11**	27.45 ± 0.05
	Avg	28.46 ± 0.25	28.49 ± 0.18	33.43 ± 0.19	28.47 ± 0.35	27.96 ± 0.24	26.54 ± 0.06	32.01 ± 1.14	25.97 ± 0.07	26.32 ± 0.10	28.57 ± 3.19	26.28 ± 0.12	26.18 ± 0.40	26.48 ± 0.17	26.28 ± 0.05	28.14 ± 0.19	32.19 ± 0.51	26.92 ± 0.15	29.24 ± 0.05	31.42	30.85	27.68 ± 0.07	31.92	29.74 ± 0.05	36.00	38.04 ± 1.02	32.32	28.62 ± 0.02	*23.13 ± 0.12*	**21.71 ± 0.08**	24.05 ± 0.02
MAPE (%)	3	30.24 ± 0.93	24.84 ± 0.51	29.71 ± 0.50	25.72 ± 0.40	27.23 ± 0.80	25.35 ± 0.64	27.74 ± 0.49	24.09 ± 0.35	25.52 ± 0.76	28.04 ± 1.67	24.99 ± 0.83	28.88 ± 3.23	25.59 ± 1.28	25.24 ± 0.63	31.60 ± 0.74	55.15 ± 2.04	27.92 ± 1.57	24.78 ± 0.36	22.56	22.43	20.71 ± 0.03	23.78	21.67 ± 0.02	23.72	24.09 ± 0.26	23.33	21.79 ± 0.02	23.40 ± 0.78	**21.39 ± 0.23**	*21.95 ± 0.18*
	6	30.30 ± 0.77	26.69 ± 0.51	29.80 ± 0.42	26.82 ± 0.39	28.24 ± 0.93	26.56 ± 0.55	29.07 ± 0.52	25.13 ± 0.23	26.46 ± 0.63	29.04 ± 1.40	26.32 ± 0.61	29.31 ± 2.63	26.69 ± 1.16	26.45 ± 0.53	33.05 ± 0.86	58.10 ± 2.24	28.96 ± 1.38	27.01 ± 0.44	25.65	25.58	22.46 ± 0.04	27.22	23.88 ± 0.03	27.23	28.45 ± 0.18	26.61	23.97 ± 0.02	23.81 ± 0.72	**21.87 ± 0.24**	*23.15 ± 0.12*
	12	32.77 ± 0.67	31.71 ± 0.65	33.40 ± 0.39	30.26 ± 0.53	30.60 ± 1.19	29.49 ± 0.46	32.30 ± 0.73	27.98 ± 0.40	29.56 ± 0.75	31.53 ± 0.96	29.50 ± 0.47	31.58 ± 2.32	29.76 ± 1.26	29.39 ± 0.45	37.84 ± 1.06	65.26 ± 2.85	31.77 ± 1.26	32.90 ± 0.72	34.04	33.94	26.38 ± 0.07	34.72	28.54 ± 0.03	34.24	35.19 ± 0.14	33.62	28.62 ± 0.03	*24.90 ± 0.70*	**22.95 ± 0.23**	25.61 ± 0.11
	Avg	30.90 ± 0.78	27.32 ± 0.53	30.68 ± 0.38	27.32 ± 0.41	28.48 ± 0.81	26.88 ± 0.52	29.42 ± 0.55	25.48 ± 0.28	26.88 ± 0.62	29.30 ± 1.38	26.66 ± 0.60	29.71 ± 2.65	27.09 ± 1.22	26.77 ± 0.51	33.77 ± 0.88	58.93 ± 2.28	29.32 ± 1.41	27.74 ± 0.47	26.75	26.63	22.84 ± 0.04	27.93	24.31 ± 0.02	27.76	28.81 ± 0.19	27.26	24.38 ± 0.02	23.94 ± 0.74	**21.98 ± 0.24**	*23.36 ± 0.11*

TFNSW

Metric	Horizon	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	Chronos2-U	Chronos2-M	Chronos2-FT-M	TimesFM	TimesFM-FT	MoE-ZS	MoE-FT	Moirai2-ZS	Moirai2-FT	STAEFormer	STAE-Ours	A2TTA
MAE	3	168.71 ± 15.14	122.66 ± 1.26	195.29 ± 1.44	143.12 ± 2.43	257.73 ± 11.31	120.27 ± 0.46	148.65 ± 3.87	114.38 ± 1.09	126.95 ± 2.36	160.08 ± 23.60	129.61 ± 0.97	124.47 ± 2.51	118.46 ± 1.42	129.39 ± 0.47	146.99 ± 8.64	132.28 ± 2.29	127.68 ± 0.63	155.32 ± 0.84	160.69	160.32	128.86 ± 0.38	190.66	165.69 ± 0.60	255.65	288.08 ± 1.27	204.29	114.89 ± 0.23	88.89 ± 1.68	**61.18 ± 0.30**	*76.26 ± 0.21*
	6	181.43 ± 14.38	145.13 ± 0.79	211.00 ± 1.58	175.19 ± 3.28	261.85 ± 11.96	142.34 ± 0.66	180.04 ± 3.54	134.84 ± 1.26	150.05 ± 2.33	177.30 ± 21.89	152.15 ± 1.41	139.81 ± 1.76	139.74 ± 2.06	147.98 ± 0.77	151.13 ± 5.35	150.92 ± 2.57	148.02 ± 0.71	191.54 ± 1.27	244.74	245.53	171.51 ± 0.85	282.31	234.82 ± 0.32	324.03	362.17 ± 0.58	301.81	151.95 ± 0.30	95.46 ± 1.73	**68.18 ± 0.34**	*88.38 ± 0.24*
	12	191.66 ± 13.98	158.22 ± 0.48	219.72 ± 1.46	203.53 ± 4.54	269.70 ± 12.03	156.34 ± 0.69	199.28 ± 4.63	148.00 ± 1.26	164.32 ± 2.37	189.11 ± 20.37	167.11 ± 1.75	154.35 ± 1.24	153.70 ± 2.03	160.78 ± 0.84	162.81 ± 4.96	170.66 ± 2.26	161.09 ± 1.47	205.55 ± 1.20	291.02	288.66	187.39 ± 0.77	345.79	256.73 ± 0.39	359.26	371.62 ± 0.68	399.37	167.54 ± 0.34	107.50 ± 1.60	**78.02 ± 0.42**	*101.46 ± 0.59*
	Avg	179.87 ± 14.50	140.15 ± 0.82	208.33 ± 1.48	170.89 ± 3.29	262.63 ± 10.87	138.04 ± 0.62	174.36 ± 3.31	130.61 ± 0.96	145.42 ± 2.41	174.19 ± 22.07	147.86 ± 1.05	137.90 ± 1.42	135.57 ± 1.80	144.41 ± 0.72	153.63 ± 6.20	148.78 ± 2.33	144.04 ± 0.82	182.20 ± 1.03	228.11	227.74	160.20 ± 0.68	266.77	215.22 ± 0.33	306.03	334.59 ± 0.89	294.05	142.36 ± 0.28	96.12 ± 1.67	**67.89 ± 0.32**	*86.78 ± 0.24*
RMSE	3	280.25 ± 20.45	217.63 ± 1.09	371.81 ± 3.45	262.49 ± 4.16	370.25 ± 14.96	215.03 ± 0.78	269.38 ± 4.49	205.28 ± 2.18	224.54 ± 4.17	292.97 ± 49.14	224.34 ± 1.36	219.33 ± 3.97	211.67 ± 3.60	229.39 ± 0.90	291.53 ± 13.80	241.64 ± 3.36	226.79 ± 1.65	277.14 ± 1.12	329.78	327.55	268.64 ± 1.59	369.29	336.90 ± 0.43	462.40	561.94 ± 2.60	417.01	252.28 ± 0.75	171.62 ± 2.96	**129.32 ± 0.65**	*154.06 ± 0.79*
	6	308.03 ± 18.86	256.45 ± 1.03	394.80 ± 3.35	326.67 ± 6.57	384.60 ± 12.76	257.56 ± 1.26	333.40 ± 6.29	244.87 ± 2.33	270.67 ± 4.71	326.33 ± 44.92	271.09 ± 2.95	250.85 ± 2.25	253.32 ± 4.77	265.94 ± 1.21	303.05 ± 8.85	280.63 ± 3.49	266.95 ± 2.18	335.83 ± 1.27	473.47	473.73	348.32 ± 2.49	519.48	458.77 ± 1.05	570.16	649.78 ± 1.17	565.48	326.33 ± 0.66	189.36 ± 3.02	**147.94 ± 0.81**	*180.12 ± 0.90*
	12	327.96 ± 18.20	282.09 ± 0.90	407.82 ± 2.97	370.26 ± 8.00	402.93 ± 11.34	283.78 ± 1.17	365.29 ± 9.55	269.48 ± 2.24	296.76 ± 4.74	346.80 ± 41.06	297.74 ± 2.86	277.73 ± 1.60	279.24 ± 4.41	290.96 ± 1.24	330.87 ± 7.88	318.01 ± 2.67	290.09 ± 2.68	358.40 ± 1.12	549.73	546.05	372.36 ± 2.11	608.15	493.26 ± 0.47	648.32	652.88 ± 0.85	692.51	352.26 ± 0.80	219.12 ± 2.63	**173.98 ± 0.61**	*211.44 ± 1.27*
	Avg	303.16 ± 19.20	247.22 ± 1.02	390.75 ± 3.25	313.61 ± 5.86	383.38 ± 12.29	247.58 ± 1.00	318.17 ± 5.87	235.35 ± 1.95	259.40 ± 4.67	318.61 ± 45.67	259.82 ± 2.21	245.35 ± 1.93	243.47 ± 4.15	257.96 ± 1.08	307.19 ± 10.11	275.25 ± 3.16	257.27 ± 2.16	318.13 ± 1.10	441.54	440.18	323.47 ± 2.06	487.06	420.65 ± 0.52	544.51	609.59 ± 1.67	545.09	303.78 ± 0.69	190.33 ± 2.83	**147.15 ± 0.64**	*176.97 ± 0.84*
MAPE (%)	3	203.98 ± 42.96	101.54 ± 2.40	167.79 ± 4.05	109.87 ± 6.56	419.02 ± 18.83	93.75 ± 3.87	121.57 ± 11.39	87.14 ± 5.01	94.78 ± 7.81	94.19 ± 7.03	134.83 ± 7.19	109.08 ± 12.15	93.03 ± 3.88	99.28 ± 3.48	112.69 ± 14.09	162.98 ± 5.57	98.77 ± 5.85	154.16 ± 5.94	59.11	60.65	54.11 ± 0.13	84.30	73.71 ± 1.57	103.01	54.54 ± 0.87	68.13	43.39 ± 0.01	59.94 ± 3.07	**49.30 ± 0.51**	*56.66 ± 1.98*
	6	210.50 ± 43.26	130.32 ± 3.10	176.75 ± 3.86	137.59 ± 9.85	427.56 ± 35.35	104.87 ± 3.92	131.86 ± 9.09	102.09 ± 3.95	108.40 ± 8.04	108.65 ± 5.53	147.27 ± 10.71	122.40 ± 16.15	105.70 ± 3.72	106.79 ± 3.16	110.90 ± 12.12	183.80 ± 8.11	114.90 ± 9.27	194.19 ± 6.66	116.26	117.43	82.43 ± 0.32	163.00	128.42 ± 0.85	164.37	164.10 ± 1.19	131.57	65.49 ± 0.27	*64.60 ± 2.35*	**53.55 ± 0.32**	66.51 ± 1.39
	12	221.79 ± 41.52	146.31 ± 4.51	188.49 ± 3.75	174.63 ± 17.60	438.74 ± 48.59	115.98 ± 3.12	137.54 ± 6.98	114.71 ± 2.98	120.47 ± 9.27	119.62 ± 3.65	162.82 ± 10.53	132.49 ± 17.32	117.09 ± 2.11	115.50 ± 2.33	111.46 ± 8.46	206.11 ± 5.57	130.80 ± 6.23	218.83 ± 4.82	180.70	182.97	99.59 ± 0.44	242.48	171.51 ± 0.37	207.96	206.18 ± 1.08	226.64	80.90 ± 0.33	*76.98 ± 1.65*	**63.01 ± 0.27**	77.64 ± 1.30
	Avg	211.72 ± 42.72	125.28 ± 3.02	177.95 ± 3.89	138.49 ± 10.57	430.83 ± 35.95	104.00 ± 3.59	129.82 ± 8.24	99.93 ± 3.54	106.92 ± 8.01	106.24 ± 4.44	148.01 ± 7.69	119.37 ± 14.85	104.21 ± 3.09	106.33 ± 3.01	112.25 ± 11.68	182.00 ± 6.10	113.97 ± 6.78	187.17 ± 5.55	119.00	120.67	78.26 ± 0.28	160.45	124.25 ± 0.83	155.80	144.74 ± 1.00	137.97	62.69 ± 0.22	66.43 ± 2.35	**54.66 ± 0.33**	*66.01 ± 1.49*